

FACTORS CONTRIBUTING TO DELAY IN DIAGNOSIS OF GULLAIN-BARRÉ SYNDROME AND IMPACT ON CLINICAL OUTCOME

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ABSTRACT: *Introduction:* Heterogeneity of presenting symptoms makes the initial clinical diagnosis of Guillain-Barré syndrome (GBS) challenging. *Methods:* Observational retrospective study from 2 teaching hospitals (Parkland Memorial Hospital and University of Texas Southwestern University Hospital) between 2008 and 2013. *Results:* Sixty-nine GBS patients were identified. GBS was suspected on initial emergency department visit in only 49%. During first hospital encounter, 58% were evaluated by a neurologist. Neuropathic pain and presence of intact deep tendon reflexes were associated with delayed GBS diagnosis ($P < 0.05$). There was significantly better clinical outcome among patients who were evaluated by a neurologist during the initial visit ($P < 0.005$). Among these patients there was also significant difference in discharge destination; 71.2% of patients evaluated by a neurologist were discharged home ($P < 0.01$). Patients in whom GBS was not suspected at the time of initial Neurology evaluation were more likely to require intubation and to have residual weakness at the time of discharge ($P < 0.05$). *Conclusions:* Atypical clinical signs and symptoms may lead to delayed diagnosis of GBS. Early neurological evaluation is associated with improved clinical diagnosis and discharge disposition.

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Guillain-Barré syndrome (GBS) can present as limb pain, numbness, weakness, paresthesias, or cranial nerve dysfunction associated with decreased or absent deep tendon reflexes (DTRs).¹ Considering the variable symptoms, initial clinical diagnosis of GBS can be challenging. The incidence of GBS is 1.72 patients per 100,000 person-years, with a 50% increase for every 10-year increase in age.² Despite the low incidence, GBS remains an essential component of the emergency department (ED) differential diagnosis of acute onset neurological deficits.³ When evaluating a patient with a neurological complaint, the ED physician must use the patient's history and physical examination to guide the differential diagnosis and to determine whether a neurological consultation is required. Early identification of GBS may lead to earlier initiation of management, including immunomodulatory therapy, inten-

sive care unit admission in selected patients, and multidisciplinary team involvement. The North American Plasmapheresis trial showed a clear benefit in the sub-group of patients in whom plasmapheresis was initiated within 7 days of onset of symptoms, emphasizing the importance of early diagnosis and prompt treatment.⁴

This observational retrospective study aims to identify clinical features that contribute to delay in the diagnosis of GBS, and to determine if early neurological consultation influences the diagnosis and outcome. We hypothesized that atypical clinical features (intact DTRs, descending or asymmetric weakness) would be associated with delayed diagnosis. We also hypothesized that neurologic consultation during initial presentation to the ED would lead to earlier consideration of GBS as the diagnosis and would positively impact clinical outcome.

METHODS

We conducted a retrospective review of electronic medical records at our 2 teaching hospitals; a public hospital for Dallas County (Parkland Memorial Hospital, PMH) and an academic referral hospital (University of Texas Southwestern University Hospital, UTSW), between January 2008 and December 2013. We also included patients who initially presented to another hospital at the time of onset of symptoms and were later transferred to PMH or UTSW. This study was approved by the UTSW Institutional Review Board and approved at both hospitals.

Potential cases were identified based on a coded diagnosis of acute immune-mediated or postinfectious neuropathy (ICD-9 code: 357.0). The cases were further narrowed to select those with a confirmed diagnosis of GBS based on the onset of progressive weakness in more than 1 limb or region and reduced or absent reflexes (DTRs) during the disease course, plus at least 1 of the following criteria: (1) bilateral weakness with strong tendency toward symmetry; (2) weakness nadir reached by 12 h to 28 days, with subsequent plateau or improvement; (3) cerebrospinal fluid (CSF) findings consistent with albuminocytologic dissociation (elevated CSF protein content >50 mg/dl and cell count <10); and (4) electrodiagnostic abnormalities consistent with

Abbreviations: CI, confidence interval; CSF, cerebrospinal fluid; DTRs, deep tendon reflexes; ED, emergency department; GBS, Guillain-Barré syndrome; IVIG, intravenous immunoglobulin; LTAC, long-term acute care; OR, odds ratio; PMH, Parkland Memorial Hospital, PMH; UTSW, University of Texas Southwestern University Hospital.

Key words: delay; diagnosis; emergency service; Guillain-Barré syndrome; referral and consultation; treatment

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GBS.^{5,6} Patients were excluded if there was evidence of another identified cause of the symptoms.

If GBS was documented in the differential diagnosis (even if not the first consideration) and some initial investigations (such as lumbar puncture) were planned, the diagnosis was considered to be correct. Study outcomes were defined by the frequency of complete/near-complete resolution of weakness (documented Medical Research Council scale strength 4+/5 or 5/5) at the time of hospital discharge, duration of hospitalization, and discharge destination.

Clinical data were analyzed using SPSS 21 software. Categorical variables were analyzed with chi-square tests. All continuous variables were analyzed using the Mann-Whitney *U*-test. *P*-values of <0.05 were considered statistically significant.

RESULTS

A total of 177 patients were screened based on ICD-9 codes, and 82 with GBS were identified. Thirteen of the patients who presented initially to another hospital were excluded due to insufficient chart information. Patients with etiologies other than GBS were excluded; these included other neuropathies such as chronic inflammatory demyelinating polyneuropathy or vasculitic neuropathy, heavy metal or drug intoxication, compressive myelopathy, spinal cord infarction, idiopathic transverse myelitis, West Nile virus infection, multiple sclerosis, acute ischemic or hemorrhagic stroke, critical illness neuropathy or myopathy, neuromuscular junction disorder, polymyositis, or severe metabolic disturbances. A total of 69 patients (an average of 11.5 per year) met the prespecified inclusion and exclusion criteria and were included in the study. Of these cases, 40 (58%) were men, and the median age was 41 years (18–70 years). Initial clinical presentation was 66.7% at PMH, 14.5% at UTSW, and 18.8% at another facility. Among the patients included in the study 43.5% were Hispanic, 31.9% Caucasian, and 17.4% African American. The most common initial symptom reported by patients was weakness in 37.7% (*n* = 26), followed by paresthesias in 24.6% (*n* = 17), neuropathic pain in 21.7% (*n* = 15), numbness in 10.1% (*n* = 7), shortness of breath in 1.4% (*n* = 1), cranial nerve involvement in 1.4% (*n* = 1), ataxia in 1.4% (*n* = 1), and autonomic dysfunction in 1.4% (*n* = 1). At the first visit to the ED, the majority of patients (50.7%; *n* = 47) had a symmetrical ascending pattern of weakness, while 11.6% (*n* = 8) had a descending pattern of weakness (proximal to distal), 10.1% (*n* = 7) had asymmetric weakness, and 27.5% (*n* = 19) had no weakness. Areflexia/hyporeflexia (absence/reduction

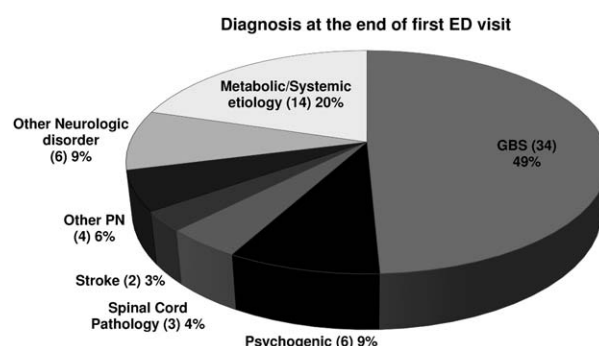


FIGURE 1. Diagnosis at the end of first ED visit. Systemic or metabolic etiologies: diarrhea (*n* = 4), dehydration (*n* = 2), respiratory tract infection (*n* = 2), CHF exacerbation (*n* = 1), renal failure (*n* = 1), acute coronary syndrome (*n* = 1), hyperglycemia (*n* = 1), bronchitis (*n* = 1), anemia (*n* = 1). Other Neuropathies: diabetic peripheral neuropathy (*n* = 1), traumatic nerve injury due to fall (*n* = 1), and in 2 patients no clear etiology of neuropathy was mentioned. Other neurological disorders: complex migraine (*n* = 3), myopathy (*n* = 1), neurological weakness, no specified etiology (*n* = 2). ED, emergency department; GBS, Guillain-Barré syndrome; PN, peripheral neuropathy.

of 4 or more reflexes) was documented in 58% (*n* = 40) patients at the initial visit.

Among the 69 patients with an ultimate diagnosis of GBS, the diagnosis was considered during the initial ED visit in only 34 (49.3%) (Fig. 1). On initial visit, GBS was suspected by the ED physician in 30.4% of patients (Fig. 2); 58% (40 patients) were evaluated by a neurologist during the initial ED visit. We found a significant association between failure to consider the diagnosis by the ED physician and a lack of evaluation by a neurologist (*P* < 0.001; odds ratio [OR], 13.50; confidence interval [CI], 2.82–64.52). Conversely, consideration of GBS as a diagnosis during the first ED visit was significantly more likely when the patient was evaluated by a neurologist (67.5%) (*P* < 0.001; OR, 8.94; CI, 2.91–27.53). Factors which were associated significantly with failure to consider GBS as the diagnosis included neuropathic pain and intact deep tendon reflexes (*P* < 0.05; OR, 0.14; CI, 0.41–0.499 and *P* < 0.05, OR, 0.07; CI, 0.01–0.35, respectively). An atypical (descending or asymmetric) pattern of weakness or no weakness was associated with lack of suspicion of GBS as the diagnosis by neurology residents (*P* < 0.05; OR, 0.25; CI, 0.09–0.74). Even after adjusting for intact reflexes and neuropathic pain on initial ED visit, there was a significant association between neurology consultation during the initial ED visit and consideration of diagnosis of GBS (*P* < 0.01; CI, 1.57–20.67; OR, 5.71).

The median number of ED visits before GBS was suspected was 2 (range: 1 to 5). Time from initial ED visit to consideration of GBS as the diagnosis was longer in patients with intact DTRs on initial visit

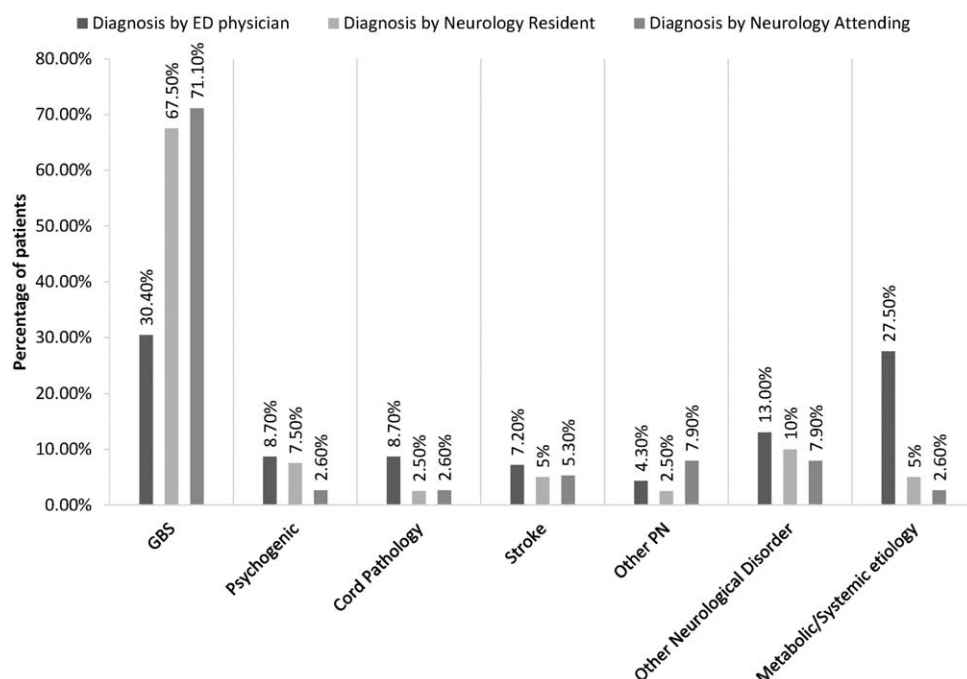


FIGURE 2. Diagnosis at the time of initial emergency department visit.

(median: 6 days vs. 1 day, $P < 0.001$). The number of ED visits until the time of suspicion of GBS was also higher for these patients ($P < 0.001$). There was no significant difference in time from initial presentation to GBS diagnosis or number of ED visits before consideration of GBS among patients presenting initially to PMH, UTSW, or another hospital.

All of the patients in this study received immunomodulatory therapy. At the time of treatment initiation 17.4% ($n = 12$) patients were ambulatory. A total of 37.7% ($n = 26$) received plasmapheresis, 42% ($n = 29$) received intravenous immunoglobulin (IVIG), and 20.3% ($n = 14$) received both IVIG and plasmapheresis. Forty-nine percent ($n = 34$) had resolution of motor symptoms at the time of discharge from the acute care setting (median length of hospitalization: 14 days). Discharge destination was home in 53.6% ($n = 37$), inpatient rehabilitation facility in 42% ($n = 29$), and long-term acute care (LTAC) in 4.3% ($n = 3$). No deaths were reported during hospitalization for GBS. The time from ED visit to initiation of immunomodulatory therapy was significantly shorter among patients who were evaluated by a neurologist compared with those who were not on their initial ED visit (median 1.5 vs. 7 days; $P < 0.005$).

Failure to consider GBS by ED physicians had a significant association with residual weakness at the time of discharge ($P < 0.05$; OR, 4.29; CI, 0.142–13.01). There was significantly better clinical outcome among patients who were evaluated by a neurologist during the initial visit ($P < 0.005$; OR, 4.87; CI, 1.72–13.81 and $P < 0.05$; OR, 3.6; CI,

1.32–9.79, respectively). There was also a significant difference in discharge destination; 71.2% of patients evaluated by a neurologist on initial visit were discharged to home (rather than an LTAC or inpatient rehabilitation facility) following treatment in the hospital compared with 27.6% of patients who were not evaluated by a neurologist on initial ED visit ($P < 0.01$; OR, 5.81; CI, 1.73–15.51). If the neurologist failed to consider GBS on initial consultation, there was an increased likelihood of residual weakness ($P < 0.05$; OR, 0.31; CI, 0.11–0.89) and intubation for respiratory failure during the hospital course ($P < 0.05$; OR, 4.37; CI, 1.23–15.48).

Residual weakness was associated with several factors (Table 1), including higher number of ED visits, increased time from initial ED visit to diagnosis, and delay in initiation of treatment. There was no significant difference in clinical outcome based on hospital of initial presentation.

DISCUSSION

This study demonstrates that the initial clinical diagnosis of GBS may be challenging; the diagnosis was not suspected during the initial ED visit in nearly 50% of patients. Lack of evaluation by a neurologist, neuropathic pain, preserved reflexes, and an atypical pattern of weakness were all associated with a delay in considering the diagnosis of GBS. The delay in diagnosis had a significant impact on outcome, as assessed by the clinical status at the time of discharge and the discharge disposition.

Table 1. Association of clinical improvement at time of discharge with diagnostic and treatment delay.*

	Clinical improvement		
	Yes	No	P-Value
No. of ED visits before GBS diagnosis	1	2	<0.005
ED visit to GBS diagnosis (days)	1	6	<0.005
Symptom onset to diagnosis(days)	7	10	0.062
Neurology resident evaluation to GBS diagnosis (days)	1	2.5	0.156
Neurology attending evaluation to diagnosis (days)	1	2	0.090
ED visit to lumbar puncture (days)	1	5	<0.01
ED visit to initiation of treatment (days)	1	6	<0.005
Symptom onset to treatment initiation (days)	6	9	0.093
Duration of hospitalization (days)	7	16	<0.001

*All durations are median values. P-values were calculated using the Mann-Whitney test. Clinical improvement defined as 4+/5 or 5/5 strength (Medical Research Council scale) at the time of hospital discharge.

ED, emergency department; GBS, Guillain-Barré syndrome.

Prompt neurological consultation not only affected the time to diagnosis and time to initiation of treatment but also had an impact on outcome variables such as resolution of weakness and discharge destination. However, due to small sample size, it was difficult to evaluate the impact of neurological consultation itself after correction for time to diagnosis and treatment.

The importance of prompt clinical diagnosis was further emphasized by the higher frequency of intubation and residual weakness when the diagnosis of GBS was not considered during initial Neurology consultation. Despite the in-house availability of a Neurology resident at all times at both hospitals, 39.8% ($n = 22$) patients with GBS were not evaluated by a neurologist during their initial ED presentation.

In a French study of 1,679 ED visits, 14% received neurological consultation, ultimately resulting in a change in diagnosis in 52.5% of cases.⁷ In another study, Moeller et al. examined the records of 493 patients with neurological consultation in ED and found that the initial ED diagnosis agreed with the consultation in 60.4% of cases.⁸ Nearly 20% of the patients had an initial diagnosis which was considerably different than the final neurological diagnosis. However, these studies did not specifically address GBS.

In 2006, McGillicuddy et al. described the challenges presented by GBS in the ED.⁹ Of 20 GBS patients evaluated in their study, 65% ($n = 13$) needed more than 1 evaluation before they were diagnosed. To address this issue, some studies have aimed at increasing ED physician knowledge, familiarity, and comfort with the GBS diagnosis.^{8,10} Others stress the importance of communication with consulting physicians and close follow-up of patients with neurological complaints.³ Our study supports these findings and conclusions and also identifies some of the clinical features that may present difficulties for treating physicians.

Like previous reports, our study has certain limitations. The number of patients was limited due to the low incidence of GBS. Also, some GBS patients may have been overlooked if the clinical diagnosis was not coded properly. We did not have long-term data on most of the patients included in the study.

This study highlights certain important clinical aspects of GBS and factors which contribute to the delay in clinical diagnosis. Awareness of atypical clinical presentations and prompt neurology consultation may facilitate timely diagnosis and management of GBS, resulting in better clinical outcomes.

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REFERENCES

- Dimachki MM, Barohn RJ. Guillain-Barre syndrome and variants. *Neurol Clin* 2013;31:491–510.
- Shui IM, Rett MD, Weintraub E, Marcy M, Amato AA, Sheikh SI, et al. 2012. Guillain-Barre syndrome incidence in a large United States cohort (2000–2009). *Neuroepidemiology* 2012;39:109–115.
- Pope JV, Edlow JA. Avoiding misdiagnosis in patients with neurological emergencies. *Emerg Med Int* 2012;2012:949275.
- Plasmapheresis and acute Guillain-Barre syndrome. The Guillain-Barre syndrome Study Group. *Neurology* 1985;35:1096–1104.
- van Doorn PA, Ruts L, Jacobs BC. Clinical features, pathogenesis, and treatment of Guillain-Barre syndrome. *Lancet Neurol* 2008;7: 939–950.
- Wakerley BR, Uncini A, Yuki N. Guillain-Barre and Miller Fisher syndromes—new diagnostic classification. *Nat Rev Neurol* 2014;10: 537–544.
- Moulin T, Sablot D, Vidry E, Belahsen F, Berger E, Lemounaud P, et al. 2003. Impact of emergency room neurologists on patient management and outcome. *Eur Neurol* 2003;50:207–214.
- Moeller JJ, Kurniawan J, Gubit CJ, Ross JA, Bhan V. Diagnostic accuracy of neurological problems in the emergency department. *Can J Neurol Sci* 2008;35:335–341.
- McGillicuddy DC, Walker O, Shapiro NI, Edlow JA. Guillain-Barre syndrome in the emergency department. *Ann Emerg Med* 2006;47: 390–393.
- Hansen CK, Fisher J, Joyce N, Edlow JA. Emergency department consultations for patients with neurological emergencies. *Eur J Neurol* 2011;18:1317–1322.